

Instructor: Dr. Rajesh Kumble Nayak

01 Sep, 2025 @ 11:00 AM.

Duration 50 Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!**Problem - 1 {15 Marks}:**In a generalised coordinate system r , the Lagrangian is given by

$$L(r, \dot{r}) = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2) - \frac{1}{2}\frac{mk}{a}(r - a)^2.$$

Here,

$$T = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2)$$

$$V = \frac{1}{2}\frac{mk}{a}(r - a)^2.$$

- Find the Lagrange equation of motion.
- Find at least one non-trivial conserved quantity.
- Is total energy E in this system.
- Find the solution to the equation of motion as an explicit function of time.

Problem - 2 {15 Marks}:Consider a Lagrangian, in a generalised coordinate system $\{\theta, \phi, \psi\}$, which is given by,

$$L = \frac{1}{2}I_1(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2}I_3(\dot{\psi} + \dot{\phi} \cos \theta)^2 - mgl \cos \theta,$$

A particle of mass m is confined to move along a vertically oriented circle under the influence of gravity. Find a generalised coordinate for this problem and write the Lagrangian for it.

- Find the constants of motion.
- Find the equation of motion.
- Is the system integrable?

Problem - 3 {15 Marks}:

Consider a free particle in a two-dimensional inertial frame,

- (1) Write the equation of motion in a plane polar coordinates.
- (2) Show that the particle moves along a straight line.